

Mechanical Waves and Sound, solved

Step-by-step solutions to every practice problem in Chapter 11. Try each problem on your own before you read its solution.

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P11.1

The **wave speed** $v = \omega/k = \sqrt{F/\mu}$ is how fast the *pattern* moves along the string. It depends only on the string's tension and mass per unit length.

The **particle speed** $\partial y/\partial t$ is how fast a given bit of string moves up and down. Its maximum is $A\omega$, which depends on the amplitude and the frequency.

Can $A\omega$ exceed v ? It would require

$$A\omega > \frac{\omega}{k} \Rightarrow kA > 1 \Rightarrow A > \frac{\lambda}{2\pi}.$$

The amplitude would have to be about one-sixth of a wavelength or more. The string would then be stretched at steep angles, and the small-amplitude assumptions behind the wave equation, $v = \sqrt{F/\mu}$ included, would break down. For ordinary waves, $A \ll \lambda$, so the particles move much more slowly than the wave.

P11.2

The speed of sound depends on stiffness divided by inertia: $v = \sqrt{\text{modulus}/\rho}$. Steel is about 6500 times denser than air, but it is around a million times **stiffer**. Its elastic modulus is about 1.6×10^{11} Pa, compared with $\gamma p \approx 1.4 \times 10^5$ Pa for air. So

$$\frac{v_{\text{steel}}}{v_{\text{air}}} \approx \sqrt{\frac{10^6}{6500}} \approx \sqrt{170} \approx 13\text{--}17,$$

depending on which modulus applies. The enormous stiffness more than wins over the higher density.

P11.3

A difference of 10 dB means a factor of $10^{10/10} = 10$ in **intensity**. So 90 dB carries ten times the power per unit area of 80 dB. (It sounds roughly twice as loud, because loudness perception is itself compressed.)

Damage to hearing depends on the energy delivered to the cochlea, which is intensity $\times$ time. Ten times the intensity delivers the same harmful dose in one-tenth of the time. That is why exposure limits shrink so quickly: 8 h at 85 dB, but only about 15 min at 100 dB. A sound that seems only “a bit louder” can be far more damaging.

P11.4

The **pitch** of the voice is set by the vibration frequency of the vocal cords, which depends on their tension and mass. Helium barely changes these, so the fundamental frequency stays nearly the same.

What changes is the **vocal tract**: the throat, mouth and nose act as resonant cavities, a bit like pipes, that amplify certain bands of frequencies called formants. Their resonant frequencies are proportional to the speed of sound. In helium, sound travels about 2.9 times faster than in air (1007 against 343 m/s), so the formants shift up strongly. The higher harmonics of the voice are then boosted relative to the lower ones. The result is a thin, “squeaky” tone, even though the fundamental pitch hardly moves.

P11.5

Compare with $y = A \sin(kx + \omega t)$:

- $A = 0.050$ m
- $k = 5\pi$ rad/m, so $\lambda = 2\pi/k = 0.40$ m
- $\omega = 50\pi$ rad/s, so $f = 25$ Hz
- $v = \omega/k = 10$ m/s, travelling in the $-x$ direction, because of the $+\omega t$
- Maximum transverse speed: $A\omega = 0.050(50\pi) = 7.9$ m/s

P11.6

Mass per unit length:

$$\mu = \rho \pi r^2 = 7800 \times \pi (1.25 \times 10^{-4})^2 = 3.83 \times 10^{-4} \text{ kg/m.}$$

Wave speed in the fundamental mode, where $\lambda = 2L$:

$$v = 2Lf = 2(0.648)(330) = 428 \text{ m/s.}$$

Tension:

$$F = \mu v^2 = (3.83 \times 10^{-4})(428)^2 = 70 \text{ N.}$$

P11.7

Light arrives almost instantly, taking about 3 μs to cover 1 km. The delay is therefore essentially the travel time of the sound:

$$d = vt = 343(3.0) = 1.0 \times 10^3 \text{ m} \approx 1.0 \text{ km.}$$

Rule of thumb. Sound covers about 1 km in 3 s ($343 \times 3 = 1029 \text{ m}$). So dividing the delay in seconds by 3 gives the distance in kilometres. In miles, divide by 5.

P11.8

(a) **At 20 m.** The distance is 10 times larger, so the intensity is 100 times smaller, which is 20 dB lower: 80 dB.

(b) **Acoustic power.** At 2.0 m:

$$I = I_0 \times 10^{100/10} = 10^{-12} \times 10^{10} = 1.0 \times 10^{-2} \text{ W/m}^2.$$

$$P = I 4\pi r^2 = (1.0 \times 10^{-2})(4\pi)(2.0)^2 = 0.50 \text{ W.}$$

Only half a watt of sound, from a machine using kilowatts of power.

(c) **Safe distance.** 85 dB is 15 dB below the level at 2.0 m, so the intensity must fall by a factor of $10^{1.5} = 31.6$. Since $I \propto 1/r^2$:

$$r = 2.0\sqrt{31.6} = 11 \text{ m.}$$

Closer than this, hearing protection is needed.

P11.9

(a) Open at both ends: $f_1 = v/2L$, so

$$L = \frac{v}{2f_1} = \frac{343}{2(65.4)} = 2.62 \text{ m.}$$

The next resonance is the second harmonic, $2f_1 = 131 \text{ Hz}$.

(b) Closed at one end: $f_1 = v/4L$, so

$$L = \frac{343}{4(65.4)} = 1.31 \text{ m.}$$

Only odd harmonics occur, so the next resonance is $3f_1 = 196 \text{ Hz}$.

A stopped pipe sounds the same note as an open pipe twice as long. That is why organs use stopped pipes for the lowest notes.

P11.10

$f \propto \sqrt{F}$, so $F \propto f^2$. To bring the higher string down from 444 to 440 Hz:

$$\frac{F_{\text{new}}}{F_{\text{old}}} = \left(\frac{440}{444} \right)^2 = 0.982.$$

The tension must be **reduced by about 1.8%**. A useful approximation:

$$\Delta F/F \approx 2\Delta f/f = 2(4/444) = 1.8\%.$$

P11.11

Two steps.

1. Wall as a stationary observer. The bat is a source moving towards it:

$$f_1 = f \frac{v}{v - v_b} = 40.0 \text{ kHz} \times \frac{343}{333} = 41.2 \text{ kHz.}$$

2. Wall as a stationary source re-emitting f_1 , with the bat as an observer moving towards it:

$$f_2 = f_1 \frac{v + v_b}{v} = 41.2 \times \frac{353}{343} = 42.4 \text{ kHz.}$$

Combined: $f_2 = f \frac{v + v_b}{v - v_b} = 40.0 \times \frac{353}{333} = 42.4 \text{ kHz.}$

The bat hears an echo about 2.4 kHz higher than its call. Bats use exactly this shift to judge their speed relative to targets.

P11.12

(a) Wavelength:

$$\lambda = \frac{c}{f} = \frac{1540}{3.5 \times 10^6} = 4.4 \times 10^{-4} \text{ m} = 0.44 \text{ mm.}$$

(b) Depth. The sound travels there and back, so

$$d = \frac{ct}{2} = \frac{1540(65 \times 10^{-6})}{2} = 0.050 \text{ m} = 5.0 \text{ cm.}$$

(c) Fat-muscle reflection:

$$R = \left(\frac{1.70 - 1.38}{1.70 + 1.38} \right)^2 = \left(\frac{0.32}{3.08} \right)^2 = 0.011 = 1.1\%.$$

Why weak echoes are useful. Because 99% of the sound passes *through* each soft-tissue boundary, the pulse can go on to produce echoes from many deeper boundaries, building up a full cross-sectional image. The receivers are extremely sensitive. The machine also applies *time-gain compensation*, amplifying later echoes more, to make up for absorption along the way. Strong reflectors such as bone and air are the real problem, because they block everything behind them.

P11.13

The frame of the pulse. Moving along with the pulse, the shape of the string is stationary and the string itself flows through it at speed v .

The top of the pulse. Approximate this part of the string as an arc of a circle of radius R , subtending an angle 2θ .

- **Length and mass:** the arc has length $2R\theta$, so its mass is $\Delta m = \mu(2R\theta)$.

- **Its motion:** it moves along the arc at speed v , so it has a centripetal acceleration v^2/R , directed down towards the centre of the circle.
- **The force on it:** the tension F pulls along the string at each end. The two pulls are tilted by θ below the horizontal, so their downward components add to $2F \sin \theta \approx 2F\theta$. Their horizontal components cancel.

Newton's second law:

$$2F\theta = \mu(2R\theta)\frac{v^2}{R} \Rightarrow F = \mu v^2 \Rightarrow v = \sqrt{\frac{F}{\mu}}. \checkmark$$

The angle θ and the radius R cancel. So every pulse shape travels at the same speed, as long as the slopes are small.

P11.14

(a) **Tension.** At height y above the lower end, the rope must support the weight of the length y below it:

$$F(y) = \mu gy.$$

$$v(y) = \sqrt{\frac{F}{\mu}} = \sqrt{gy}.$$

The pulse speeds up as it climbs. Remarkably, the result does not depend on μ .

(b) **Travel time:**

$$t = \int_0^L \frac{dy}{v} = \int_0^L \frac{dy}{\sqrt{gy}} = \frac{2\sqrt{L}}{\sqrt{g}} = 2\sqrt{\frac{L}{g}}.$$

(c) **Numbers:**

$$t = 2\sqrt{\frac{5.0}{9.8}} = 1.4 \text{ s.}$$

P11.15

(a) Blood speed:

$$v = \frac{c \Delta f}{2f_0 \cos \theta} = \frac{1540(2000)}{2(4.0 \times 10^6)(0.707)} = \frac{3.08 \times 10^6}{5.66 \times 10^6} = 0.54 \text{ m/s.}$$

(b) Error. $v \propto 1/\cos \theta$, so

$$\frac{dv}{d\theta} = v \frac{\sin \theta}{\cos \theta} = v \tan \theta \Rightarrow \frac{\delta v}{v} = \tan \theta \delta \theta.$$

(c) A 5° error is $\delta \theta = 0.0873$ rad.

ANGLE	$\tan \theta$	ERROR IN v
30°	0.577	5%
60°	1.73	15%
80°	5.67	50%

Guideline. The same uncertainty in angle gives a small error at modest angles but a huge error near 90° , where $\cos \theta$ changes very rapidly. Keeping the angle at or below about 60° keeps velocity errors to a clinically acceptable level. This matters, because decisions such as grading a carotid stenosis depend on the measured peak velocity.

P11.16(a) Superposition. Use $\sin(a - b) + \sin(a + b) = 2 \sin a \cos b$ with $a = kx$ and $b = \omega t$:

$$y = 2A \sin kx \cos \omega t.$$

- Nodes where $\sin kx = 0$: $kx = n\pi$, so $x = n\lambda/2$. - Antinodes where $|\sin kx| = 1$: $x = (n + \frac{1}{2})\lambda/2$.

(b) The vibrating string. In the third harmonic, $L = 3\lambda/2$:

$$\lambda = \frac{2(1.20)}{3} = 0.80 \text{ m}, \quad v = f\lambda = 150(0.80) = 120 \text{ m/s.}$$

The nodes are spaced $\lambda/2 = 0.40$ m apart, at $x = 0, 0.40, 0.80, 1.20$ m. The fundamental is $f_1 = 150/3 = 50$ Hz.

P11.17

(a) **Mach angle:**

$$\sin \theta = \frac{1}{M} = 0.50 \Rightarrow \theta = 30^\circ.$$

(b) **Time to hear the boom.** The cone trails behind the jet. The observer hears the boom when the surface of the cone reaches them. At that moment, the jet is a horizontal distance x beyond the overhead point, where

$$\tan \theta = \frac{h}{x} \Rightarrow x = \frac{h}{\tan 30^\circ} = \frac{10 \text{ km}}{0.577} = 17.3 \text{ km}.$$

The jet flies at $2(340) = 680$ m/s, so

$$t = \frac{17\,300}{680} = 25 \text{ s}.$$

(c) It has travelled **17 km** past the observer.

The boom arrives long after the jet has passed overhead. That is why the sound seems to come from an empty patch of sky behind the aircraft.

P11.18

(a) **Distance.** Both waves travel the same distance d , and the S-waves arrive 20 s later:

$$\frac{d}{3.5} - \frac{d}{6.0} = 20 \text{ s} \Rightarrow d(0.2857 - 0.1667) = 20 \Rightarrow d = \frac{20}{0.119} = 168 \text{ km}.$$

A useful rule: each second of S-P delay corresponds to about 8.4 km.

(b) **How many stations?** One station gives only the distance, so the source could lie anywhere on a circle around it. Two stations give two circles, which cross at two possible points. A **third** station picks out the right one, locating the epicentre on the surface. This is *trilateration*, the same principle as GPS. Finding the depth as well requires additional stations and arrival-time data.

(c) **The liquid outer core.** S-waves are shear (transverse) waves, and they need a medium with shear stiffness. Liquids have zero shear modulus, so S-waves cannot travel through them. After a large earthquake, seismometers more than about **104°** around the Earth from the epicentre record P-waves but no direct S-waves. This S-wave *shadow zone* is explained only if the Earth has a large liquid region at its centre that blocks S-waves. Beno Gutenberg located the boundary of the

core, about 2900 km below the surface, in 1913. Harold Jeffreys showed in 1926 that the outer core must be liquid.

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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